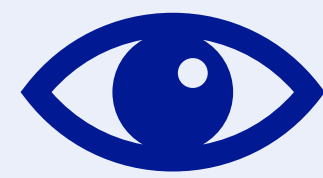


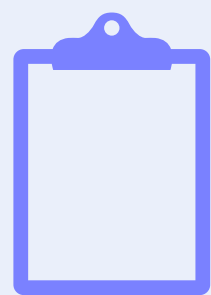
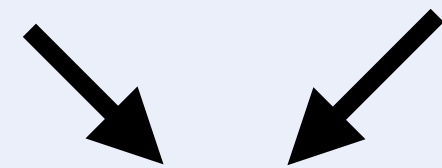
## Eye Data Collection



Retinal  
Imaging



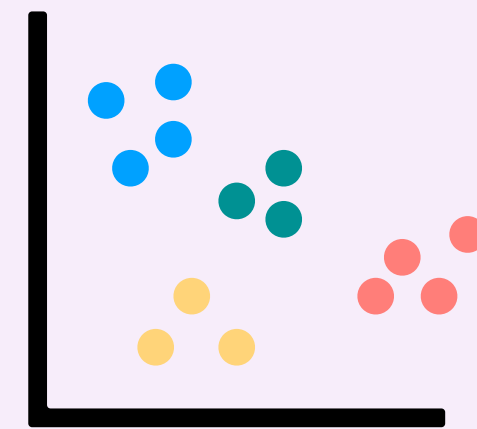
Visual  
Assessment



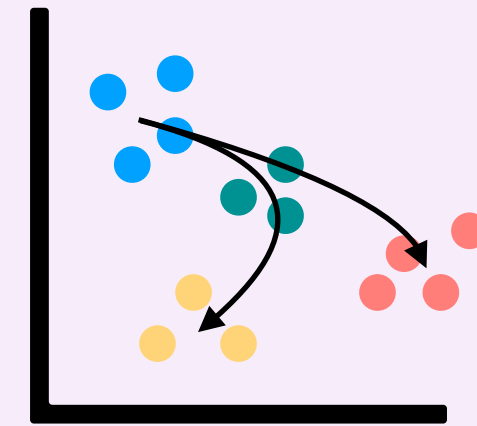
Clinical Phenotypes

- 87 Clinically graded eye phenotypes
- 7,888 Eyes (3,971 patients)
- 2,822 Patients genotyped

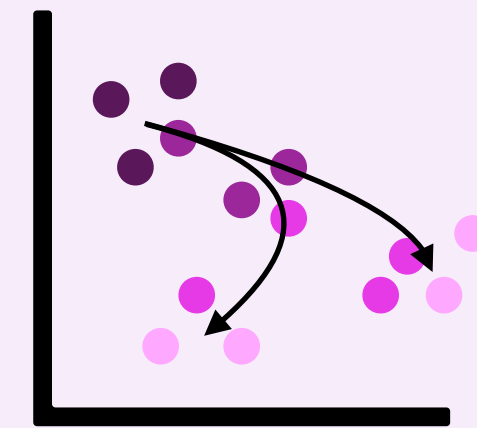
## scRNA-seq Style Analyses



Eye Clustering



Differential Disease  
Progression Trajectory



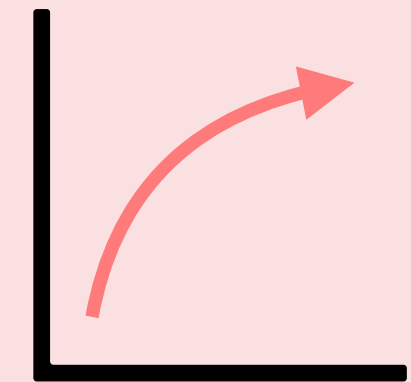
Disease  
Pseudotime

- 12 Clinically distinct clusters of disease severity
- Holistic and data-driven progression score
- Found two differential progression trajectories (neurodegenerative and vascular)

## Investigative Analyses



Trajectory Risk Factors



Long-term  
Progression Pattern

- T2D and age associate with neurodegenerative route
- *Mir9-2* and *PHGDH* variants affect progression trajectory
- Non-linear long-term progression rate